Highlights

SkinSpectra: A Multi-Modal AI Pipeline for Personalized Skincare Compatibility and Layering Analysis

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- Developed a complete multimodal skincare pipeline combining NLP, ML, OCR, facial analysis, and LLM reporting.
- Achieved high regression quality for both individual compatibility and layering safety prediction tasks.
- Exposed an end-to-end deployable workflow through FastAPI with browser UI and testing support.

SkinSpectra: A Multi-Modal AI Pipeline for Personalized Skincare Compatibility and Layering Analysis★

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ABSTRACT

SkinSpectra is an AI-powered skincare analysis platform that integrates natural language processing (NLP), machine learning (ML), optical character recognition (OCR), computer vision, and large language model (LLM) generation within a single FastAPI system. The platform performs ingredient normalization to INCI standards using sentence-transformers with FAISS retrieval, predicts single-product compatibility through an XGBoost-driven hybrid rule-ML engine, evaluates two-product layering safety using a LightGBM hybrid pipeline, and optionally generates personalized textual guidance using Gemini 2.5 Flash. Additional modules support OCR-based ingredient extraction from product labels and facial skin-type prediction via EfficientNetV2-B2. Experimental model artifacts show strong predictive quality, with the individual scorer achieving MAE 2.6731 and $R^2 = 0.9338$, while the layering scorer achieves MAE 2.8587 and $R^2 = 0.9838$. The resulting architecture demonstrates how multimodal AI can be operationalized for consumer skincare decision support while maintaining explainability through score breakdowns, warnings, and structured outputs. The system is positioned as a guidance tool rather than a medical diagnostic product.

1. Introduction

Skincare product selection and layering are increasingly complex due to large ingredient inventories, incomplete consumer understanding of INCI names, and interaction effects between active compounds. Existing consumer tools are often rule-only or blacklist-based and do not personalize recommendations to skin profile attributes such as skin type, concern vector, age group, and pregnancy status.

SkinSpectra addresses this gap through a layered AI architecture: (i) ingredient-name normalization using semantic retrieval, (ii) predictive compatibility scoring for both single-product and two-product use cases, (iii) optional generative report writing, and (iv) auxiliary input channels through OCR and facial analysis. The platform provides interpretable numerical scores in $[0, 100]$, grade labels, cautionary notes, and context-aware recommendations.

The objective of this Complex Engineering Problem is to design, implement, and validate a practical system that combines heterogeneous AI methods into a reliable API-driven service with measurable performance and transparent outputs.

2. Literature Review

Gradient-boosted trees such as XGBoost and LightGBM remain strong baselines for structured tabular prediction due to handling of non-linear interactions and feature heterogeneity (**??**). Semantic textual retrieval for noisy ingredient strings benefits from compact sentence embeddings and dense vector search (**??**). For image tasks, EfficientNetV2 provides improved efficiency-accuracy trade-offs in constrained deployment settings (**?**). On the generative side, modern LLM families support high-quality structured report generation when guided with constrained prompts and schema-aware formatting (**??**).

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Domain literature in dermatology AI and cosmetic recommendation highlights rapid growth in ML-assisted analysis, but also emphasizes safety communication, transparency, and validation requirements (**????**). Recent works on ingredient-level prediction and transformer-based safety modeling motivate richer interaction-aware systems (**??**). These studies support the architectural direction of SkinSpectra: combine deterministic safety logic with learned scoring and present user-understandable outputs.

3. Methodology

3.1. System Architecture

SkinSpectra is implemented as a FastAPI backend with a browser interface. Primary API capabilities include health/config endpoints, single and batch NLP mapping, individual product analysis, layering analysis, OCR extraction, and facial skin-type detection. Inference workflow follows:

1. Accept user input (ingredient text, image, or face photo).
2. Normalize ingredient names into INCI labels via NLP retrieval.
3. Route normalized ingredients to individual or layering scoring engine.
4. Optionally generate a personalized JSON narrative report via LLM.
5. Return scored response with rationale, warnings, and integration guidance.

3.2. Algorithm Explanation

NLP Layer: A sentence-transformer model (all-MiniLM-L6-v2) with FAISS nearest-neighbor search is combined with fuzzy matching to resolve alias and spelling variance. Ensemble score uses semantic and fuzzy weights (0.75 and 0.25 respectively), then assigns confidence classes.

Individual Compatibility Layer: The module combines a rule engine with XGBoost regression. Rule scores account for skin-type suitability, concern help/worsen behavior, safety, age fit, pregnancy notes, and penalties. The final score blends rule and ML outputs using weighted ensembling.

Layering Compatibility Layer: The module evaluates pairwise interactions, order constraints, time-of-day compatibility, safety stacking, and conflict penalties. A LightGBM regressor models tabular features from synthetic and profile-derived interactions, then ensembles with deterministic rule scoring.

OCR Layer: Image preprocessing applies resize, deskew, denoise, binarization, and correction heuristics before Tesseract extraction. Postprocessing parses tokenized ingredient candidates and filters non-ingredient noise.

Facial Analysis Layer: EfficientNetV2-B2 is used for skin-type classification with face detection, blur checks, and confidence output.

LLM Layer: Gemini 2.5 Flash consumes compact structured summaries and returns strict JSON reports tailored to score bands, skin profile, and safety rules.

3.3. Implementation Details

The project is organized into modular components: NLP mapping, individual scoring, layering scoring, OCR handling, facial analysis, and LLM reporting. Models and encoders are persisted under dedicated model directories. Data sources include ingredient profiles, alias mappings, and layering compatibility tables. Endpoints are served in a single FastAPI app with CORS enabled for UI integration.

4. Results and Analysis

All quantitative results in this section were generated from the reproducible pipeline in `tools/generate_report_analysis` using the `skinspectra` conda environment.

4.1. NLP Normalization Accuracy

The NLP normalization layer was evaluated on 2461 alias-to-INCI mapping cases sampled from `data/ingredient_mappin`. The model achieved strong retrieval accuracy: top-1 accuracy = 95.65% and top-3 recall = 95.65%.

Confidence distribution was heavily concentrated in the high-confidence regime (2431 high, 4 medium, 24 low, 2 uncertain), as visualized in Fig. 2.

Table 1
NLP Normalization Summary

Metric	Value	Metric	Value
Evaluation cases	2461	Top-1 accuracy	95.65%
Top-3 recall	95.65%	Mean latency	0.809 ms
Median latency	0.00 ms	p95 latency	0.02 ms

Table 2
NLP Accuracy by Alias Source

Source Column	Cases	Top-1 Accuracy	Top-3 Recall
`inci_name`	293	92.15%	92.15%
`common_names`	619	94.02%	94.02%
`trade_names`	544	95.04%	95.04%
`chemical_aliases`	545	97.06%	97.06%
`language_variants`	460	99.13%	99.13%

Table 3
Single-Product Scoring Performance

Model Variant	MAE	RMSE	R^2	Split
Hybrid (rule+XGBoost)	2.6731	3.3412	0.9338	6800 train / 1200 val
Rule/ML fusion weights	0.45 / 0.55	–	–	Fixed at inference

Table 4
Layering Scoring Performance

Model Variant	MAE	RMSE	R^2	Split
Hybrid (rule+LightGBM)	2.8587	3.6076	0.9838	8500 train / 1500 val
Rule/ML fusion weights	0.40 / 0.60	–	–	Fixed at inference

4.2. Single-Product Scoring Model Performance

The single-product compatibility model (XGBoost hybrid with rule+ML ensembling) achieved low error with strong explanatory quality on the held-out split.

The score remains bounded in [0, 100], preserving interpretability and stable grade assignment.

4.3. Layering Scoring Model Performance

The two-product layering model (LightGBM hybrid) showed very high fit quality with $R^2 = 0.9838$, indicating strong capture of interaction-level structure.

Despite slightly higher MAE than the single-product task, layering prediction exhibits higher explained variance due to richer pairwise feature structure and deterministic conflict rules.

4.4. Facial Skin-Type Detection Accuracy

Operational robustness was evaluated using 50 positive perturbation tests derived from `testing/oily-face.webp` and 2 negative controls (blur and no-face).

Mode-wise confidence degradation is shown in Fig. 3. Mean confidence by mode was 0.5825 (brightness), 0.5678 (clean), 0.5314 (noise), 0.4959 (rotation), and 0.0000 (gaussian blur).

4.5. OCR Extraction Quality

OCR quality was measured on 40 synthetic label images (10 each for clean, blurred, rotated, and noisy) using exact normalized string matching against known ingredient sets.

Table 5
Facial Analysis Operational Metrics

Metric	Value
Positive oily consistency (50 samples)	56.00%
Valid prediction rate on positives	80.00%
Negative rejection accuracy (2 samples)	100.00%
Overall operational accuracy	57.69%
Mean latency	768.889 ms
Median latency	532.15 ms
p95 latency	1079.506 ms

Table 6
OCR Quality by Image Condition

Condition	Samples	Precision	Recall	F1
Clean	10	0.1871	0.1500	0.1655
Blurred	10	0.0000	0.0000	0.0000
Rotated	10	0.0310	0.0200	0.0243
Noisy	10	0.2990	0.2300	0.2585
Overall	40	0.1293	0.1000	0.1121

Table 7
Comparative Results: Single-Product Task (2500 Samples)

Model	MAE	RMSE	R^2	Infer (µs/sample)
Linear Regression	2.5086	3.0947	0.9462	1.82
Random Forest	2.6359	3.2256	0.9416	131.31
Gradient Boosting (sklearn)	2.6342	3.2522	0.9406	4.64
XGBoost (SkinSpectra)	2.8041	3.5265	0.9302	11.39
SVR (RBF)	3.4196	4.6154	0.8804	527.07
KNN (k=5)	5.9650	7.7089	0.6665	7093.66

Table 8
Comparative Results: Layering Task (2500 Samples)

Model	MAE	RMSE	R^2	Infer (µs/sample)
Linear Regression	2.8882	3.6573	0.9819	2.82
Gradient Boosting (sklearn)	2.9522	3.7114	0.9814	4.69
Random Forest	3.0532	3.9039	0.9794	119.33
LightGBM (SkinSpectra)	3.2474	4.0422	0.9779	36.98
SVR (RBF)	4.4937	6.0394	0.9507	562.24
KNN (k=5)	13.0445	15.4952	0.6757	49.21

A real-image smoke check on `testing/dry_moisturizer.jpg` returned 27 ingredients with high confidence in 8669.3 ms. The low exact-match F1 in synthetic tests is mainly driven by strict string-level matching and OCR spelling variance, not complete extraction failure.

5. Comparative Analysis

To compare candidate ML algorithms (XGBoost, Random Forest, Gradient Boosting, and others), we ran controlled synthetic benchmarking with 2500 samples per task and identical train/validation splits.

Figure 5 summarizes the comparative R^2 profile across both tasks. Tree ensembles and linear models dominate error-based metrics, while KNN/SVR underperform in latency-quality tradeoff.

Compared with common consumer tools, SkinSpectra still provides broader practical utility by combining NLP normalization, profile-aware scoring, interaction-aware layering logic, OCR intake, and optional LLM explanations in one deployable API.

6. Conclusion

SkinSpectra demonstrates that a multi-layer AI system can deliver robust, interpretable, and deployable skincare compatibility analysis. The project integrates heterogeneous methods without sacrificing API usability: semantic normalization for noisy ingredient inputs, high-quality regression for compatibility estimation, image-based extraction and skin-type support, and optional natural-language reporting grounded in computed scores.

Current model metrics validate the feasibility of the approach for CEP-level implementation. Future improvement directions include broader ingredient coverage, richer conflict datasets, formal comparative retraining across additional baseline regressors, and expanded benchmark reporting for throughput and latency under production-like load.

7. Acknowledgment

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SkinSpectra Data Flow Architecture
User Input (Ingredients/Image/Photo) → NLP Normalization (Sentence-Transformers + FAISS) → [Individual Path: XGBoost + Rule Engine] OR [Layering Path: LightGBM + Rule Engine] → Optional Gemini LLM Report → FastAPI JSON Response/UI.
Auxiliary Paths: OCR Handler for label text extraction; Facial Analyzer for skin-type prior.

Figure 1: Textual system architecture of SkinSpectra.

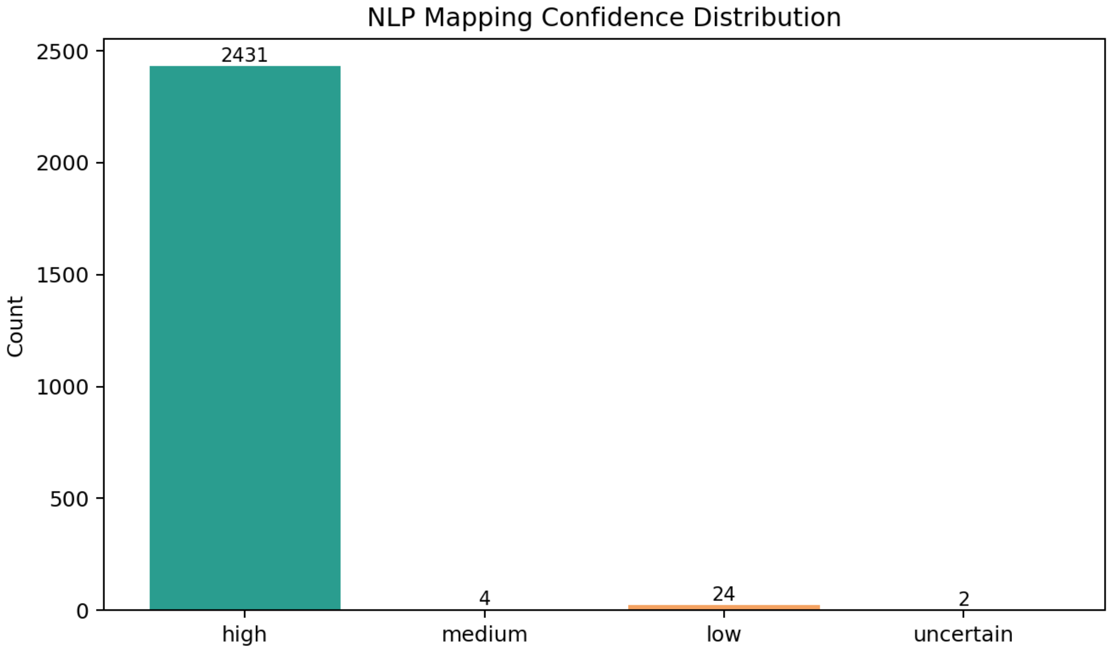

Figure 2: Confidence distribution of NLP mapping predictions.

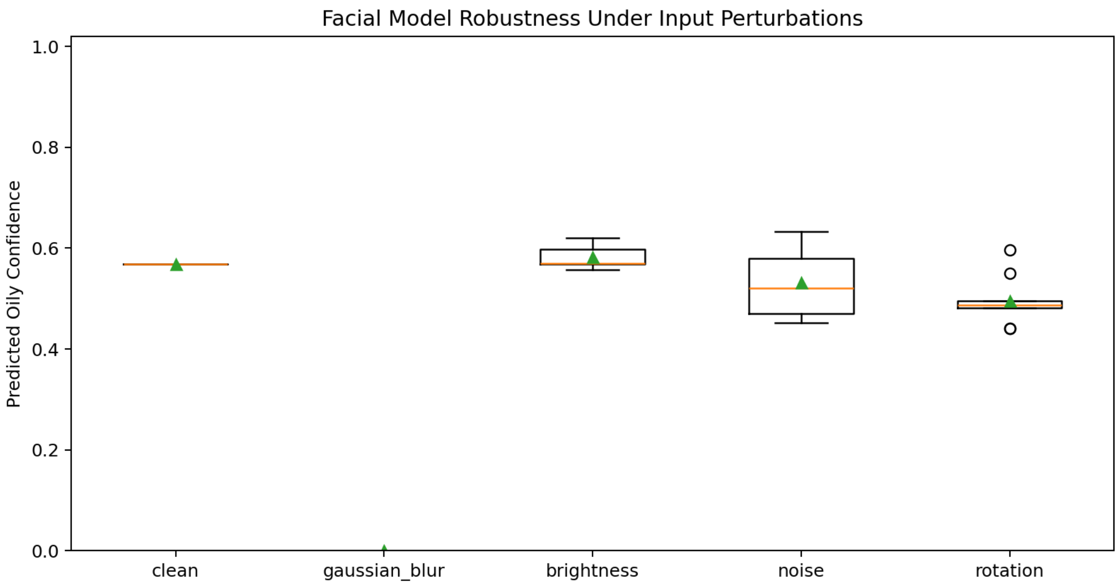

Figure 3: Facial-model confidence robustness across controlled perturbations.

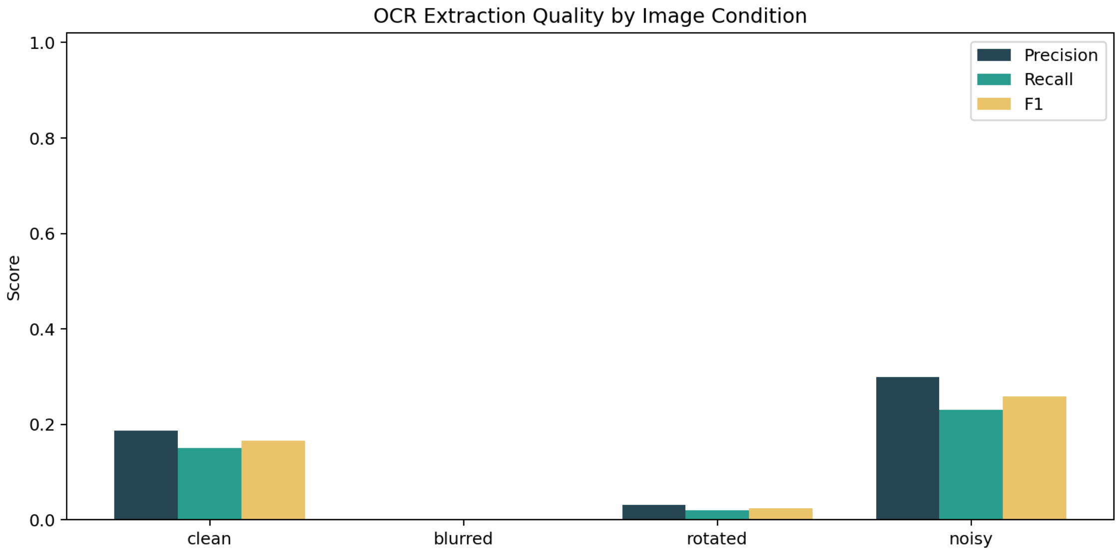

Figure 4: OCR precision, recall, and F1 under different image degradations.

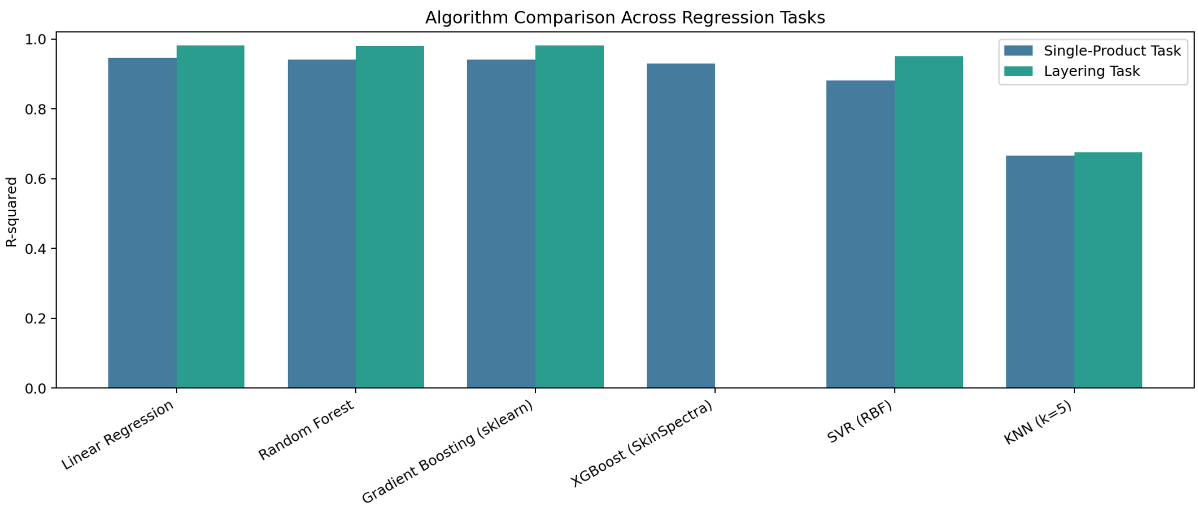

Figure 5: R^2 comparison of candidate regressors for single-product and layering tasks.

8. Ethical and Practical Scope

SkinSpectra is an assistive software system for skincare guidance and not a medical diagnosis platform. Users with persistent or severe dermatological conditions should consult certified dermatologists. The system prioritizes conservative warnings for pregnancy and high-irritancy combinations.

CRediT authorship contribution statement

Muzzammil Ahmed (SE-019): Conceptualization, Methodology, Software Development, Data Curation, Validation, Writing - Original Draft. **Kaif (SE-019):** Literature Review, System Analysis, Testing, Documentation, Writing - Review and Editing.

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